

Yoshita Lakshmi Mitta

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PROFESSIONAL SUMMARY

Computer Science student and Generative AI Developer with hands-on experience building LLM-powered, conversational, and real-time AI applications using Python, LLMs, Gemini, RAG, prompt engineering, and Voice AI. Experienced with FastAPI, Node.js, REST APIs, and AI-driven workflows. Built and deployed practical GenAI systems involving conversational interactions, document intelligence, and AI-powered decision-making.

EDUCATION

Rajiv Gandhi University of Knowledge Technologies (IIIT RK Valley)
Bachelor of Technology – Computer Science (CGPA: 9.2)

Kadapa, AP, India
2023 – 2027

SKILLS

AI & GenAI: Large Language Models (LLMs), Generative AI, Conversational AI, Prompt Engineering, Retrieval-Augmented Generation (RAG), Embeddings, Semantic Search
Programming: Python, JavaScript, C++, SQL
FullStack: FastAPI, Node.js, Express.js, REST APIs, API Integration, React
Databases: PostgreSQL, MongoDB, Supabase, Qdrant
AI/LLM Tools: Gemini API, LangChain, Nomic Embeddings
Developer Tools: Git, GitHub, Postman, Chrome DevTools

PROJECTS

AI Brain — Document Intelligence System | [Github](#)

- Tech:* React.js, Node.js, Express.js, MongoDB, LangChain, Gemini 2.5 Flash, Qdrant, Nomic Embeddings
- Designed and developed an end-to-end **LLM-powered Retrieval-Augmented Generation (RAG)** pipeline for document ingestion, PDF text extraction, section-wise chunking, batch embedding generation, vector indexing, and context-aware question answering.
- Built a **hybrid retrieval** pipeline combining semantic vector search and keyword search, improving retrieval relevance by **40%** through optimized embedding and retrieval strategies.
- Developed scalable backend APIs with conversational memory and a modular architecture for document processing, semantic retrieval, and multi-session AI workflows.

AuraMom – AI-Powered Maternal Wellness Platform | [GitHub](#) | [Live-Demo](#)

- Tech Stack:** React.js, FastAPI, Gemini API, Hume AI TTS, Supabase
- Built a **personalized maternal wellness platform** delivering **pregnancy-stage-specific nutrition guidance**, **AI-powered meditation sessions**, and **daily wellness recommendations**.
- Engineered an **end-to-end AI pipeline** leveraging **Gemini API** and **Hume AI TTS** to analyze **user mood**, dynamically generating **context-aware meditation scripts** and **natural voice-guided audio sessions**.

CareRoute AI — Real-Time AI Emergency Dispatch Engine | [GitHub](#)

- Tech Stack:** React.js, Node.js, Express.js, Supabase, LiveKit, Leaflet.js, Google Gemini 1.5 Flash
- Developing an **agentic AI emergency-response platform** that uses a voice-based AI doctor for symptom collection, preliminary triage, and downstream hospital and ambulance coordination.
- Built a **real-time voice AI pipeline** using **LiveKit WebRTC**, and **Gemini** for low-latency patient-AI conversations and intelligent symptom analysis.
- Designed a **multi-constraint hospital-matching and resource-reservation workflow** considering hospital availability, distance, medical resources, and emergency requirements.
- Designed **real-time ambulance GPS tracking**.

EXPERIENCE

Web Development Mentor

Sep. 2024 – Nov. 2024

Student Recreation Center (SRC)

- Mentored 30+ students in frontend development and explained technical concepts through interactive coding sessions, strengthening communication and problem-solving skills.